



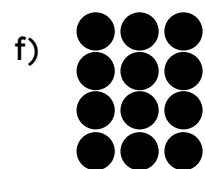
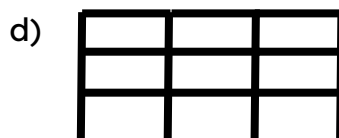
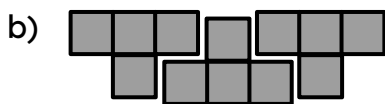
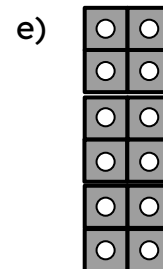
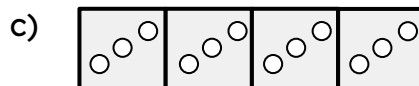
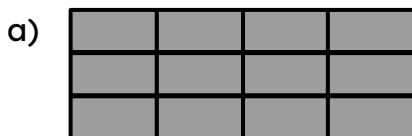
Formalising a sequence

Task A: Pictorial sequences

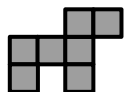
1) Andeep is making different patterns that go up in fours.

Alex is making different patterns that go up in threes.

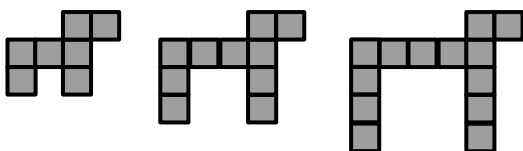
They show one image from each of their patterns. Which are Andeep's, which are Alex's and which could be either?



Sam has drawn a 'dog' by shading in squares. Their classmates each continue the pattern in different ways.

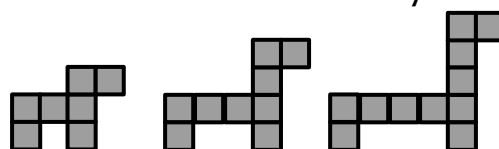


2) Aisha continues it this way



a) Describe how the pattern is developing in words

3) Lucas continues it this way

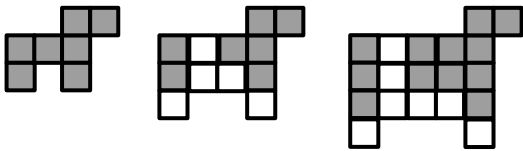


a) Describe how the pattern is developing in words

b) How many squares would be needed for the 10th dog?

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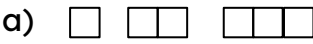
4) Izzy continues it this way



- a) Describe how the pattern is developing in words
- b) How many squares would be needed for the 10th dog?

Task B: Describing patterns in pictorial sequences

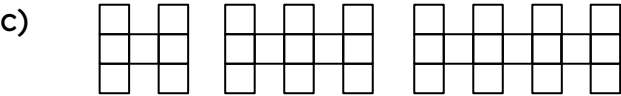
1) For each sequence, fill in the table for the number of squares and perimeter. Describe the rule for both sequences.



Term	1	2	3	4	5
Squares	1	2			
Perimeter	4	6			

Rule for squares:

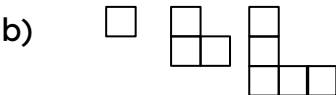
Rule for Perimeter:



Term	1	2	3	4	5
Squares	7				
Perimeter					

Rule for squares:

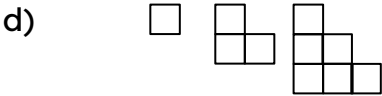
Rule for Perimeter:



Term	1	2	3	4	5
Squares	1				
Perimeter	4				

Rule for squares:

Rule for Perimeter:

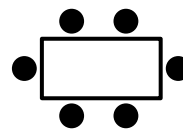


Term	1	2	3	4	5
Squares					
Perimeter					

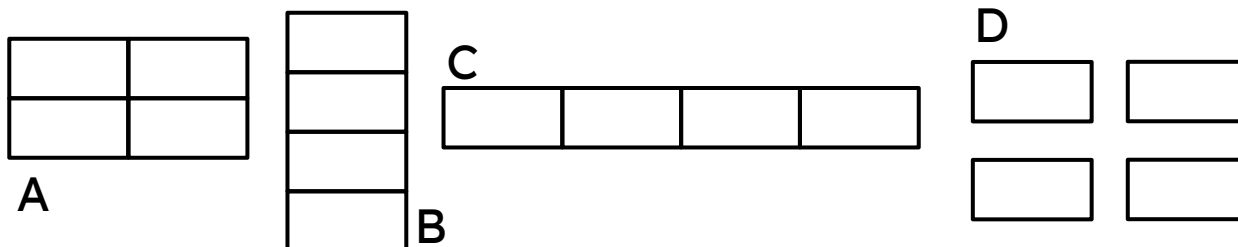
Rule for squares:

Rule for Perimeter:

2) In a school dining hall, 6 chairs (circles) fit around one rectangular table



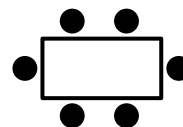
Below are 4 ways to arrange 4 tables



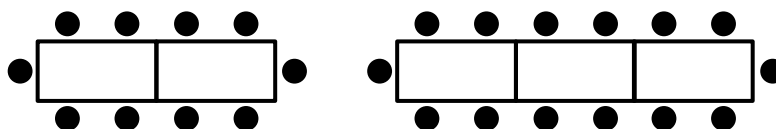
a) To calculate the number of chairs needed, the caretaker multiplies the number of tables by 6. Which way are they planning on arranging the tables?

b) For each arrangement how many more chairs can be added if an extra table is added?

3) In a school dining hall, 6 chairs (circles) fit around one rectangular table



The tables are arranged like this



a) How many chairs will fit round 4 tables?

b) What is the relationship between the number of chairs and the number of tables?

c) How many chairs can fit round 10 tables?

d) Jacob has 25 classmates. How many tables are needed so all 26 of them can have a chair.

